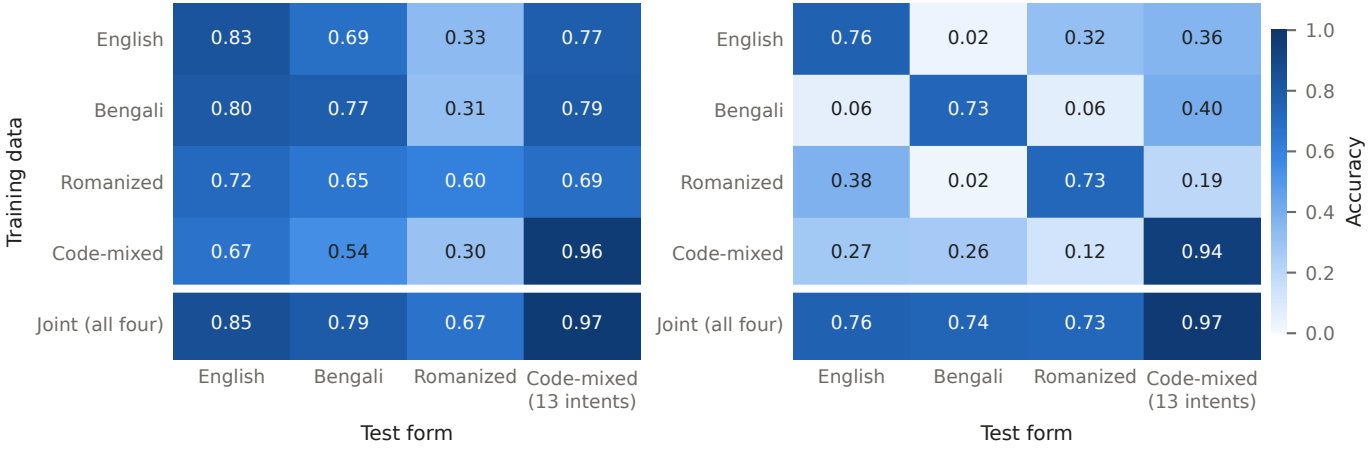
(a) Decision model (Qwen3.5-0.8B)
(b) Char n-gram TF-IDF baseline
Training data
English
Bengali
Romanized
Code-mixed
Joint (all four)
0.83 0.69 0.33 0.77
0.80 0.77 0.31 0.79
0.72 0.65 0.60 0.69
0.67 0.54 0.30 0.96
0.85 0.79 0.67 0.97
0.76 0.02 0.32 0.36
0.06 0.73 0.06 0.40
0.38 0.02 0.73 0.19
0.27 0.26 0.12 0.94
0.76 0.74 0.73 0.97
English
Bengali
Romanized
Code-mixed (13 intents)
Test form
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0